The energy that becomes thermal

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Account for mechanical energy changes through heating and deformation.

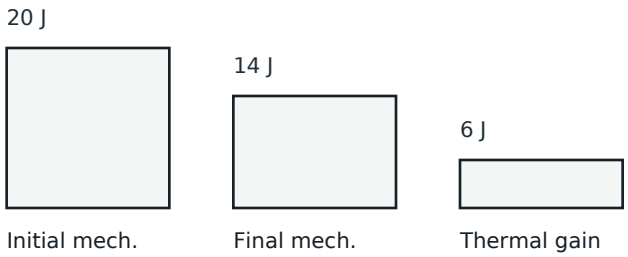
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 15.*
75 J; spring energy scales with compression squared.

02. Worked example: explain the model
Opposing arrows; mechanical energy decreases; box-floor thermal energy increases under stated assumptions.

03. Read and interpret the model
Yes; 20=14+6.

Model / diagram check



04. Apply the physics
Work -15 J; thermal gain 15 J if no other energy changes.

05. Explain your reasoning
No; mass, material/specific heat, phase and losses matter.

06. Review: connect an earlier idea *Review: Unit 4, Day 6.*
No. Work and kinetic-energy change must refer to the same endpoints. Measure matching intervals.

07. Exit check
2 J; must seek another transfer/storage or measurement error.

08. STRETCH: test the model
Distance 5 m; thermal increase 20 J if friction is the only energy transfer.

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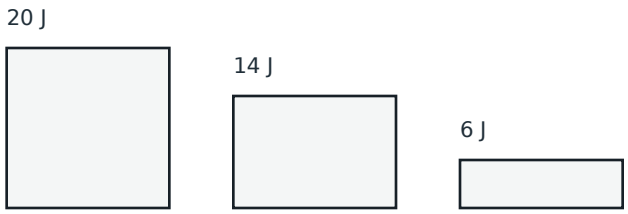
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 15.*
$|x|=\sqrt{(2\ U/k)}=0.05$ m; $|F|=15$ N.

02. Worked example: explain the model
Opposing arrows; mechanical energy decreases; box-floor thermal energy increases under stated assumptions.

03. Read and interpret the model
Yes; 20=14+6.

Model / diagram check



04. Apply the physics
25-4=21 J; internal increase 21 J.

05. Explain your reasoning
No; mass, material/specific heat, phase and losses matter.

06. Review: connect an earlier idea *Review: Unit 4, Day 6.*
W=triangle area=2.4 J; $v_f \approx 3.464$ m/s.

07. Exit check
9 J; 6 N.

08. STRETCH: test the model
Height 2 m; gravity work 60 J, friction -20 J; K 40 J; $v=\sqrt{(80/3)}\approx 5.16$ m/s.